KUSHAL KRISHNAPPA

kushalkrishnappa333@gmail.com | Boston, MA | +1 (857) 351-3884 | kushalkrishnappa.com
linkedin.com/in/kushalkrishnappa | github.com/kushalkrishnappa

PROFESSIONAL SUMMARY

Software Engineer with 2.5+ years experience building distributed systems across on-prem and cloud environments, from OpenStack fleet orchestration for 9,000+ VMs at Pure Storage to multi-cloud security pipelines at RoonCyber.AI. Experienced in Go, Python, Kubernetes and event-driven architecture, with a focus on reliability and performance at scale.

EDUCATION

Northeastern University — *MS in Computer Science* (GPA: 4.0/4.0) Jan 2025 - May 2027
Courses: Design Paradigms, Algorithms, Computer Systems, Agentic AI *Boston, MA*

PES University — *B.Tech in Computer Science and Engineering* Aug 2018 - Jun 2022
Courses: Machine Learning, Network & Information Security, Big Data, Operating Systems *Bengaluru, India*

TECHNICAL SKILLS

- **Languages:** Go, Python, Java, TypeScript, SQL
- **Backend & Distributed Systems:** gRPC, REST APIs, Temporal, Concurrency, Async I/O, Multi-Tenancy
- **Cloud & Infrastructure:** AWS, Kubernetes, Nomad, Consul, Docker, Terraform, Ansible, Packer, Linux, OpenStack
- **Databases & Messaging:** PostgreSQL, SQLAlchemy, MongoDB, Redis, RabbitMQ
- **CI/CD & Observability:** Jenkins, GitHub Actions, Nginx, Prometheus, Grafana, OpenTelemetry, RCA
- **Developer Tools:** Claude Code, MCP, Git, Visual Studio, IntelliJ, Jira, Confluence

PROFESSIONAL EXPERIENCE

RoonCyber.AI — *Software Engineer (Co-op), Boston, MA* Jan 2026 - Aug 2026

- Took the AWS-hosted platform multi-cloud with **Azure posture detection**: Lambda collector, **event-driven** normalization into the findings schema and APIs for cross-cloud risk and blast-radius analysis.
- Built a **multi-tenant M2M API auth system** (key lifecycle, Postgres RLS tenant isolation, caching/RBAC/rate-limiting middleware), unblocking enterprise deals requiring programmatic access.
- Extended the eBPF security pipeline with custom header-scanning rules that isolate the offending field in flagged HTTP requests, cutting triage time for application-layer incidents.
- Built an **8-stage agentic dev harness** driving Claude from PRD to merged PR with human validation at each gate, helping team ship 25+ features over 3 months and now in daily use by 9+ engineers.

Everpure (Pure Storage) — *Member of Technical Staff 2, Bengaluru, India* Jan 2023 - Dec 2024

- Built an event-driven reconciliation service (Python, Temporal) end-to-end driving failed testbeds back to known-good state, replacing manual triage and cutting **MTTR from over 5 hours to 10 minutes** across a 2,000-testbed OpenStack fleet (9,000+ VMs); scoped to service-account events so operator-initiated offlines were never clobbered.
- Replaced the Jenkins-based recovery path with snapshot-restore workflows (Go, Temporal) feeding the same reconciliation service and **cutting recovery time by further 85%**.
- Designed and built the **fleet-disruption-manager service** end-to-end gating disruptive image rollouts, with per-group concurrency budgets in **Consul** so one bad image could not drop a 3,000-machine test pool and rollouts could be canaried with fast revert; later **ported to Go and open-sourced** in Cloud-Foundations/Dominator.
- Cut database reads by **∼7x** under polling from **thousands of agents** with a 100s-TTL **Redis cache** and a **negotiated backoff coefficient** in the upstream client.
- Turned **Ubuntu 14 to 22 migration of dev VM fleet** into permanent release gate; reconciled divergent team workflows into one Packer golden image, beta in 1 month, then automated the corpus as a pytest suite replacing manual QA.

Mercedes-Benz R&D — *Software Engineer, Bengaluru, India* Aug 2022 - Dec 2022

- Built a multi-source ingestion service (Java, Spring Boot) feeding **Azure Event Hubs** behind a generic interface for heterogeneous payloads and the PySpark/Delta Lake pipelines processing **90TB+** of telemetry into OTA rollout analytics.
- Drove **p95 latency down 50%** on the OTA service's hottest endpoints during on-call by identifying an N+1 query pattern.

PROJECTS

- **Auth Server** | Built a multi-tenant auth service in **Go** using **gRPC** and **Protocol Buffers**, implementing JWT-based auth, token rotation, tenant isolation and low-latency inter-service communication with focus on scalability and security.
- **Beacon** | Designed and built a **notification service in Go using Temporal**, enabling reliable event-driven email alerts across 2 personal web services; set up SMTP server with Cloudflare DNS and exposed a REST API for service integration.
- **dsapanicle.com** | Built and shipped algorithmic challenges platform using **AI-driven development (Claude Code, agentic workflows)**, serving 10-20 active users with full CI/CD via GitHub Actions + Hostinger webhooks.

PUBLICATIONS

An Enhanced Deployment of 5G Network Using Multi-Objective Genetic Algorithm - IEEE, 2022 | Implemented a **genetic algorithm to optimize dense 5G deployment**, minimizing cost while maximizing coverage and efficiency.

Certifications: Hashicorp Terraform Associate (004), AWS Fundamentals Specialization, Web Application Security Testing with OWASP ZAP, Object Oriented Programming - Advanced (Java OOP), Foundations of Responsible AI Learning